Ankush Kumar Mallick

Noida, India | +91 9142623599 | ankushmallick750@gmail.com | linkedin.com/in/ankush-mallick750 | github.com/Ankush-22

Professional Summary

Computer Science graduate specializing in Machine Learning and applied NLP/LLM systems. Builds end-to-end ML pipelines (data cleaning, feature engineering, hyperparameter tuning, model evaluation) and Retrieval-Augmented Generation (RAG) applications using LangChain, Llama 3.1 (via Groq API), and vector databases (FAISS, ChromaDB). Strong foundation in Data Structures & Algorithms and DBMS; takes a build-from-scratch approach — implementing core logic instead of relying solely on high-level abstractions.

Technical Skills

Languages: Python, C++, C, SQL
ML/DL Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost
ML/DL Concepts: Supervised & Unsupervised Learning, ANN, CNN, RNN, Transformers, Regression, Classification
ML/DL Techniques: Random Forest, Hyperparameter Tuning, Cross-Validation, Feature Engineering
NLP & LLM: LangChain, RAG (Retrieval-Augmented Generation), Hybrid Search (BM25 + Vector), TF-IDF, NLTK, HuggingFace Embeddings, Sentence-Transformers, Groq API, Llama 3.1, Prompt Engineering
Vector Databases: FAISS, ChromaDB
Data Libraries: Pandas, NumPy, Matplotlib, Seaborn
Web/Backend Frameworks: FastAPI, Flask, Streamlit
Databases: PostgreSQL, SQL
Core CS: Data Structures & Algorithms (DSA), OOP, DBMS
Tools: Git, GitHub, Linux Bash, Jupyter Notebook, VS Code

Projects

YouTube Video RAG Assistant | github.com/Ankush-22/YouTube-Video-RAG-Assistant

- Built a multi-video RAG research assistant that ingests transcripts from multiple YouTube URLs and answers cross-video questions, eliminating manual video review.
- Designed a semantic retrieval pipeline using HuggingFace embeddings (all-MiniLM-L6-v2) indexed in FAISS, paired with Llama 3.1 (8B) via the Groq API for low-latency response generation.
- Implemented automatic topic extraction and one-click summary generation (concise, detailed, study-notes modes) through an interactive Streamlit chat interface.

CiteMind – Multi-Document RAG Assistant | github.com/Ankush-22/CiteMind

- Developed a multi-document RAG application supporting PDF, DOCX, and PPTX ingestion for citation-backed Q&A across heterogeneous document sets.
- Engineered a Hybrid Search pipeline combining semantic vector search (ChromaDB + HuggingFace embeddings) with BM25 keyword search, fused via Reciprocal Rank Fusion (RRF) for higher retrieval accuracy.
- Used Llama 3.1 (Groq) to generate answers with in-line source-filename citations, exposing retrieved chunks in the UI for full source transparency.

Travel Budget Prediction System | *Python, XGBoost, Scikit-learn, Plotly, Streamlit*

- Built an ML regression model to predict travel budgets using feature engineering on inputs such as travel style and trip duration.
- Applied numeric preprocessing and hyperparameter tuning (Scikit-learn/XGBoost), achieving a 0.97 R^2 score on test data, with Plotly-based visual analytics.
- Deployed as an interactive Streamlit app for real-time forecasting, with joblib model serialization and a requirements.txt for reproducibility.

Education

University of Engineering & Management, Jaipur **Aug 2022 - Jun 2026**
Bachelor of Technology in Computer Science Engineering *Rajasthan, India*
- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming

Certifications

Python for Data Science — **NPTEL**
The Joy of Computing using Python — **NPTEL**